Data Engineer Intern Assignment

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1. Executive Summary

The project analyzes Airbnb listings in New York City using data engineering, statistical analysis, machine learning, and AI techniques. The goal is to understand pricing behavior, demand patterns, and market segmentation.

Key findings:

- Entire homes are significantly more expensive than private rooms
- Location is a major driver of pricing differences
- Superhost status strongly improves review scores
- A small group of professional hosts controls large portions of supply
- Machine learning models explain ~37% of price variation

Business impact:

- Supports dynamic pricing strategies
- Helps identify profitable neighborhoods
- Enables investment decision-making
- Provides AI-driven market insights

2. Objectives & Scope

Objectives

- Analyze Airbnb NYC market using multiple analytical techniques
- Build predictive models for pricing
- Understand demand, reviews, and host behavior
- Develop AI-driven insights system

Scope

- Cities: New York City (primary analysis)
- Optional: London, Paris (cross-city comparison framework)
- Focus on listings, calendar, reviews, neighbourhood data

Prioritization rationale:

- Focused deeply on NYC due to dataset richness
- Ensured strong statistical + ML + AI coverage over quantity of cities

3. Dataset Overview

- listings.csv.gz → 35,036 rows, 91 columns
- calendar.csv.gz → 12.7M rows
- reviews.csv.gz → 1M+ rows
- neighbourhoods.csv → 230 rows

Relationships

- listings.id → calendar.listing_id
- listings.id → reviews.listing_id

Limitations

- Scraped data (not official Airbnb database)
- Missing values in optional fields
- Estimated revenue is not real financial data
- Calendar shows availability, not confirmed bookings

4. Methodology

The analysis followed a structured pipeline:

I. Data Cleaning
II. Feature Engineering
III. Exploratory Data Analysis (EDA)
IV. Statistical Hypothesis Testing
V. Machine Learning Modeling
VI. NLP Analysis (reviews)
VII. Recommendation System Design
VIII. AI-driven insight generation

Reasoning:

- Combines descriptive + predictive + generative analytics
- Ensures both statistical rigor and business interpretability

5. Engineering Approach

Raw Data → Cleaning → Feature Engineering → Master Table → Analysis Layer → ML Models → AI Layer

Key decisions:

- Used pandas for ingestion (simplicity)
- Used DuckDB for SQL analytics layer
- Built modular pipeline structure
- Avoided heavy distributed systems (not needed at dataset scale)

Trade-offs:

- Not fully production-grade (no Airflow)
- Prioritized flexibility over scalability

6. EDA Findings

- Price distribution is highly skewed
- Entire homes dominate premium pricing
- Manhattan/central areas show higher prices
- Seasonal demand patterns exist
- Professional hosts dominate supply

Business interpretation:

- Market is segmented into budget vs luxury
- Location is strongest pricing factor

7. Statistical Findings

- Entire homes vs private rooms → significant difference ($p < 0.001$)
- Superhosts have higher ratings (large effect size)
- Neighbourhood explains ~7% of price variance
- Review volume negatively correlates with price

Interpretation:

- Pricing is structurally driven by property type + location
- Reputation impacts perceived quality
- Demand does not always increase price

8. Data Science Experiments

Model	Performance
Linear Regression	baseline
Random Forest	best performance
Gradient Boosting	competitive

- Best model explains ~37% variance in price
- Bedrooms and accommodates are strongest predictors
- Location adds major explanatory power
- Non-linear models outperform linear ones

9. AI/ML Experiments

NLP

- Sentiment analysis of reviews
- Topic modeling (cleanliness, location, host behavior)

LLM Applications

- AI assistant for Airbnb insights

- Automated report generation
- Pricing advisor system

Recommendation System

- Content-based similarity using cosine similarity
- Nearest-neighbor optimization for scalability

10. Visualizations

- Price distribution histogram
- Heatmap of correlations
- Boxplots by room type
- Neighbourhood price comparison
- Regression scatter plots

Explain:

- Visuals confirm skewed distribution
- Clear segmentation between property types
- Strong location effects

11. Business Recommendations

- Use dynamic pricing for entire homes
- Focus expansion in high-demand neighbourhoods
- Improve listing quality to increase ratings
- Encourage Superhost adoption
- Optimize pricing based on seasonality

12. Cross-City Comparisons

If used NYC + London + Paris:

- NYC → highest volatility and pricing spread
- London → regulated and stable pricing

- Paris → seasonal tourism-driven demand

Insight:

- Different cities require different pricing strategies
- One global model is not sufficient

13.Limitations & Caveats

- Scraped dataset limitations
- Missing booking history
- No user-level interaction data
- Model explains only part of price variation
- Sentiment model is basic NLP

14.Future Improvements

- Deep learning models for pricing
- Transformer-based sentiment analysis
- Real-time pricing system
- Full RAG-based AI assistant
- Production deployment using cloud + Airflow

15.Reflection

This project required balancing depth and scope. Prioritization was made toward building a complete end-to-end system rather than focusing on a single advanced model.

Trade-offs:

- Chose interpretability over complexity
- Focused on NYC instead of too many cities
- Built modular but not fully production infrastructure

Key learning:

- Data engineering is as important as modeling
- Business interpretation is critical
- AI adds value when combined with analytics

Appendix A: AI Usage Disclosure

Tools

- ChatGPT (GPT-4/5)
- Python (pandas, sklearn)
- statsmodels
- spaCy

AI-assisted sections

- Code generation
- Report structuring
- Statistical interpretation

Validation

- All outputs checked manually
- Statistical tests verified
- Models evaluated using cross-validation